Luke Conibear

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Overview

Generalist engineer specialising in simplifying complex scientific software systems at the intersection of software engineering, AI, cloud infrastructure, and atmospheric science.

Experience

Senior Software Engineer, Tomorrow.io, US (Remote) | 2022–present

- Primary contributor to a department-wide cloud cost efficiency initiative, delivering a 25% spend reduction in 2025 through information-content compression (`xbitinfo`), spot/reserved capacity strategy, KEDA scale-to-zero, rightsizing, lower cross-cloud/cross-region transfer, and stricter data-retention policies.
- Designed, deployed, and maintained production microservices and data pipelines in Python, Rust, and Go for ML and physics-based forecasting workflows, operating within a highly asynchronous distributed team (YouTube).
- Core contributor in Severity-1 incident response; identified recurring failure modes and led preventive fixes that reduced MTTR by 35% from 2024 to 2025.
- Led design and delivery of multi-cloud MLOps infrastructure across Azure ML, AWS SageMaker, and GCP Vertex AI.
- Led simplification and modernisation of legacy forecasting applications, improving reliability, maintainability, and operational resilience.
- Led department-wide adoption of AI-assisted engineering workflows, integrating code review standards, clear documentation, and CI/CD improvements to increase delivery speed and code quality.
- Own and evolve an event-driven distributed system (Go) coordinating multi-cloud weather data pipelines across AWS, Azure, and GCP using PostgreSQL-backed dependency tracking.

Research Software Engineer Machine Learning, University of Leeds, UK (Remote) | 2021–2022

- Taught AI emulators of numerical atmospheric models.
- Authored and maintained an open-source course for High-Performance Python and Machine Learning.

Research Fellow, University of Leeds, UK (Remote) | 2018–2021

- Engineered AI emulators (`scikit-learn`, `tpot`) to act as fast proxies for complex atmospheric simulations (~20 TB of data), distributing training across high-performance compute clusters (`dask`).
- Maintained and supported `WRFotron`, an open-source automation tool for atmospheric models, managing GitHub releases, documentation, and a community of users.

Education

- **PhD**, Ambient Air Quality and Human Health, University of Leeds | 2015–2018
  - Computational models & applied data science
- **MSc**, Bioenergy, University of Leeds | 2014–2015
- **BEng**, Mechanical Engineering, University College London | 2007–2010

Publications

- Lead-author of 9 academic papers (h-index of 19 with 1,600+ citations).

Tools

- **AI-native:** Cursor, Codex, GitHub Copilot.
- **Languages:** Python (scientific & ML focus), Rust, Go, Bash.
- **Cloud:** Azure (ML, AKS, storage, event grid), GCP (Vertex AI, GKE, GCS, Pub/Sub), AWS (SageMaker, S3).
- **Platform & Ops:** Git (GitHub), Kubernetes, Docker, Terraform, Argo, GitHub Actions, GPUs, DataDog, PagerDuty, DoIT, Jira.